

**PIPING GENERAL
SPECIFICATION**

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1 GENERAL OVERVIEW

1.1 Scope

This specification covers the design and layout of all process and aboveground utility piping for the Karabatan Power Plant project of KUS, located in KARABATTAN, Atyrau region, Kazakhstan.

1.2 Codes, Standard and Specification

The major applicable codes, standards and specification are those listed below:

ASME	American Society of Mechanical Engineers
ASTM	American Society for Testing and Materials including C533, C512 and C159 Thermal Insulation
AWWA	American Water Works Association

If more codes are in contrast, the most stringent shall be use. Local regulations if more stringent will be applied.

1.3 Units of measurement

Geometric and physical measures shall be expressed as units of measurements of the INTERNATIONAL SYSTEM (SI). The piping diameter shall be expressed in INCHES.

1.4 Plot Arrangement

The general layout shows the arrangement of major features, such as tanks, buildings, roads and operating units.

Individual unit plot plans (equipment arrangement) show the arrangement of equipment and structure within the operating unit.

1.5 Stream Data Sheet

This document is issued during the process engineering phase with the purpose of identifying, through a reference shown on the Piping and Instrumentation Diagrams, lines or lines systems. It contains the following information necessary for the piping detailed design development: type of fluid, basic piping material, operating and design conditions,

hazardousness class, extent and type of non-destructive tests, piping classes and heat treatments.

1.6 Line List

It is a document complementing the P&I Diagrams for the piping development, with information on: fluid identification, line number, material class, supplementary indication, “coming from-going to” or service of line or equipment item for line of equipment piping appurtenance, P&I diagrams references, references for type of running of the line, operating and design temperature and pressures, welding class, heat treatment, line testing, hazardous of fluid, equipotent connections, painting cycle, nominal pipe size, insulation thickness, of each individual line.

1.7 Loads and supports

The piping systems and their supports shall be designed to resist the effect of loads imposed by the weight of the pipe live and dead, thermal, dynamic, settlement, movement, impact, wind, snow, earthquake, valves, fittings, insulation and fluid in the lines.

All permanent line supports shall be designed to satisfactory withstand loads caused by the hydrostatic test of the lines, in addition to the ones acting during the normal operating conditions.

Other dynamic loads, such as the ones caused by relief valve opening, vibration caused by high pressure drops across control valves or by a sudden valve closures, may require special supports and shall be taken into account in the design of piping systems.

The stresses created by the imposed loads shall not exceed the allowable stresses prescribed by the ASME Code for Pressure Piping or, where applicable, the ASME Boiler and Pressure Vessel Code.

2 LAYOUT

2.1 Clearances

The requirement of 11000-PI-SPC-0002 shall be followed.

2.2 Foundation Heights

Column and vertical vessel foundations are normally 250 mm above high point of paving. Specific foundation heights may be adjusted due to grade.

The foundation height for small pumps may be increased to accommodate the 600 mm pump shaft centreline. The foundation height for large pumps should not be reduced below the standard 250 mm. In order to improve block valve accessibility, operating platforms may be applied. However, if the block valves do not pose accessibility limitations, and the process requirements are satisfied, the shaft centerline may exceed the prescribed height range.

A uniform height of support pedestal for a group of exchangers or equipment shall be used whenever practical. Set the foundation height using the exchanger, which has the greatest distance between exchanger centerline and bottom of lower makeup piping.

2.3 Piping Layout

The thermal cycle and the plant BOP (Balance Of Plant) piping systems shall be designed in accordance with the standard ANSI B31.1 Power piping.

Process lines, utility lines instrument ducts shall be on overhead pipeways at established elevation (inside the buildings).

To facilitate the maintenance of piping and the equipment they are connected to, the following actions shall be taken:

Allow complete inspection;

Keep clear passages and accesses;

Keep clear space for maneuvering and supporting dismantled parts;

Keep clear spaces for maneuvering hoist and bridge cranes;

Provide adequate flanged connections to permit removal of equipment or parts thereof;

Provide a support system suitable for keeping in position the piping and cut-off devices that are not to be dismantled.

Minimum piping elevation at floor level or above the walking planes shall be 500 mm.

The underground piping shall be adequately protected against corrosion.

Passage between units areas shall be kept free from piping at grade.

Pockets shall be avoided in piping arrangements wherever possible.

3 MAINTENANCE FACILITIES

3.1 Davits and Monorails

Use of crane shall be minimized where davits and monorails can be provided.

3.2 Machinery

The arrangement of machinery shall provide for lifting down the machinery components into a drop area at grade by mobile equipment or monorail with hoist. Machinery piping shall be arranged to minimize pipe removal during maintenance operations.

Overhead cranes, hoists, trolleys, etc, necessary for the safe and efficient operation and maintenance of the plant will be provided for the following areas:

- Electric motor driven overhead travelling (bridge) crane for the turbine hall;
- Overhead hoists/trolleys for pumps (boiler feed water, closed cooling water pumps, etc) other than they can be reached by forklift;

3.3 Vessel nozzle access

All vessel nozzles will have platform access where required for normal operation. For instrument take off points and sample connections, safe maintenance access shall be provided.

3.4 Control Valves

All control valves shall be easily accessible for operation and maintenance. For easy removal of Control Valve, a minimum of 300 mm of free space above the control valve should be available.

4 PIPE, FLANGE AND FITTINGS

4.1 Piping Materials

The Piping Materials Classes specify the piping component type and execution, basic material and schedules.

4.2 Flanges

The Piping Material Classes identify the flanges by type, material, rating and facing. Provide flanges only at equipment connections, valves, and other necessary locations. Use flat face flanges with full faced gaskets against cast iron flat face valves, equipment with flat face nozzle and against flanges in fibreglass piping systems.

Provide drop out spool in piping that must be removed frequently for maintenance or for installation or removal of temporary suction strainers and blanks.

4.3 Direction changes

Changes in direction shall be made with fittings or bends. Miter bends constructed of pipe may be used for low-pressure process and for auxiliary piping, except boiler feed water.

ELBOWS: all buttwelding elbows shall have a minimum radius of 1.5 the nominal diameter of the pipe unless otherwise specified in the isometric drawings.

LONG RADIUS ELBOWS: they shall be used only where required by piping spec where the usage and the required radius are specified. All these elbows shall be made from seamless pipe.

4.4 End Closures

Use welding caps to close headers or dead ends of lines where no future expansion is contemplated. Where future extension is anticipated or clean out is necessary, close the line with a blind flange. Threaded plugs or caps shall be used for threaded end closures.

5 EXPANSION

5.1 General

Piping displacement strains (flexibility) analyses are based on the design conditions of the lines, characterized by the most severe temperature condition sustained during various plant phases (e.g. shut-down, anomalous conditions etc.).

5.2 Loops, bends and joints

Where applicable, route the pipe to permit normal bends and offsets to take the expansion. Where this is not sufficient, provide expansion loops. Install preferably expansion loops in a horizontal positions only. Movement in horizontal lines at changes in direction shall be limited to avoid interference with adjacent lines. Where possible avoid the use of expansion loops in flare header and relief valve tail pipes. These lines require special drainage and support considerations. Use expansion joints where space limitations or pressure drop prohibits the use of piping offsets to absorb expansion.

Each expansion joint shall have description of preliminary tension (mm) in the specification and isometrics

5.3 Anchors

Use the minimum number of anchors to properly direct the movement of the pipe. Use standard anchors, where applicable. Use rigid frame pipe supports for high thrust anchors or multiple anchor locations.

5.4 Struts

Use struts in place of standard anchors and guides where structural steel is not available. They can aid in overcoming support friction without putting undue forces on the nozzles of sensitive equipment like compressors and pumps. Struts are preferred over standard anchors or guides in piping subject to mechanical or fluid induced vibration, impact or dynamic loads.

6 ALLOWABLE LOADS

6.1 General

The allowable forces and moments on nozzles of machinery and sensitive equipment will be specified in the appropriate purchase specifications by the manufacturer. As a minimum requirement, most manufacturers shall use the following rules in determining the allowable nozzle loads.

6.2 Pumps

The allowable loads on centrifugal pumps with steel or alloy cases shall meet the load criteria of API 610 or Manufacturer's standard.

6.3 Vessels and heat exchangers

The allowable nozzle loads shall be in accordance with TM 300.1/03E.

7 EQUIPMENT

In particular cases it is allowed to consider nozzle flexibility in order to reduce thrusts caused by thermal loads.

7.1 Plate Type Heat Exchanger piping

The P&IDs show valves, instrument connections, and the direction of flow through the exchangers. Design the piping to provide a removable section of the exchanger and block valves, so that full access is available for bundle pulling, plates for maintenance. Design the piping to provide wrench room for unbolting exchanger flanges.

7.2 Pumps

Pump drives may be located toward the pipe way. Pumps handling hydrocarbons at any temperature shall not be located under air cooler.

Arrange pump suction lines to be as short and direct as possible. Provide flexibility and support so that minimum pipe load is transmitted to the pumps. Support pump piping from overhead when possible. If piping must be supported from grade, use adjustable pipe supports. Do not support from maintenance dropout spools.

Whenever possible, locate the valves in the pump piping so they are accessible for hand operation. Where pump discharge piping goes to an overhead pipe way, install the block and check valves in the vertical piping above the pump. Arrange the piping on side suction pumps, and double suction (internally) pumps, so that the final elbow leading to the nozzle allows balanced flow to the impeller.

Eliminate all vapour pockets in pump suction lines. Any reduction in line size of a horizontal suction line at the pump nozzle is made through an eccentric reducer with the top flat.

Provide permanent strainer between the suction line block valve and the pump. Check to be sure that the removable pipe spool at the pump suction will permit the installation and removal of the suction strainer.

Suction piping to pumps handling liquids at or close to their vapour pressure required special care or suppresses flashing. Provide as much vertical drop as possible at the suction source before starting the horizontal run to the pump.

7.3 Storage Tanks

Provide steel block valves directly against all lower tank nozzles. One exception, if the P&IDs shown a branch connection between the tank nozzle and block valves the blocks shall be located as near as practicable to the tank nozzle. All lines are to be designed to account for settlement. Note on the piping assembly drawings that permanent piping shall not be connected to the tank shell until after the completion of hydrostatic testing of the tank.

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Aboveground tank piping for any tank or group of tanks enclosed by a dike shall not run through the dike area of any other tank or group of tanks. Tank piping shall run as directly as possible to the outside of the dike area.

Pumps shall be located outside the dike area unless a heavy viscous stock required the pump to be located within the dike area to limit the suction line pressure drop.

Other equipment including shell tube exchangers shall be located outside the dike area.

Piping connected to tanks shall be designed in such a way draining point will completely drain all media in the tank.

The storage shall be surrounded by bounded area capable of retaining 110% of the contents of on tank.

8 INSTRUMENT PIPING

8.1 General

This section covers the installation of instrument connections that, for design purposes, are best shown on the piping arrangement and detail drawings. This includes level gauges, displacements type level controllers, float type level switches, pressure gauges, orifice meter piping, temperature connections, and rotameters.

The piping instrument connections shall be in accordance with the piping material class for each particular service.

8.2 Level instruments

LEVEL SWITCHES: position the level switches in a safe accessible location from grade or platform for maintenance.

LIQUID LEVEL GAUGES: position the level gauges, so that they are visible from a common operating aisle. The normal operating level shall be visible from grade or platform. Provide platform or safe working ladder for access. Where bridle block valves are provided at the vessel differential expansion between cold bridle and hot vessel must be considered.

LEVEL CONTROLLERS: orient the level controller so that the corresponding level gauge is visible from operators. Provide a safe working platform for access.

8.3 Pressure Instruments

Local gauges and pressure transmitters shall be accessible from floor, grade or platform; they should be rigidly installed on the lines to which they are connected. The temperature taps on horizontal piping shall be located:

On piping top for gas, steam; On one side for liquids.

8.4 Flow Instrument

Orifice flanges may be placed in horizontal or vertical runs. When orifice plates are used in vertical pipe in wet vapour or saturated steam service, the flow must be downward. Provide

the pipe diameter upstream and downstream of orifice flanges in accordance with Instrument specification. Provide sufficient clearance at orifice flanges for installation of instrument piping and seal pots where required. Orientation of orifice flange taps shall be in accordance with Instr. Spec. The electrical continuity shall be guarantee between the in-line instrument and the relevant pipe.

Locate, show and identify all inline flow indicators on the piping drawings. Orient indicators to allow a good view for operators.

8.5 Temperature Instrument

Thermowells and pressure gauge at heat exchangers necessary for operating reasons shall not be located on exchanger nozzles. Place them in the piping near the exchanger.

Locate all temperature points on columns (vessel, reactors converter etc.), so that they are accessible form a ladder or platform.

The temperature taps shall be located on pipe top if installed on horizontal piping.

8.6 Local Mounted Instruments

The item number and approximate location of all local mounted instruments shall be shown on the piping drawings.

9 PIPING SYSTEMS AND SERVICES

9.1 Air Piping

Design the air piping for good line drainage. Provide drain valves at the low points of the system. Branch connections are taken off from the top of the headers.

9.2 Steam Piping

All branch lines off the steam header are taken off the top of the line. If a block valve is required, place it as close as possible to the header.

Design the steam piping for complete condensate removal. Provide condensate pots with steam traps at the low point of the system for header 2 INCHES and larger.

Headers in size 1 ½ INCH and smaller shall be arranged to drain condensate directly to a trap. It is recommended that traps have to be spaced about every 30000mm.

Condensate piping is designed for steam service.

9.3 Steam Traps

Steam traps shall be provided for the removal of condensate from collection points in live and exhaust steam systems. These collection points include condensate pots, steam separators, convectors, unit heaters, and steam tracing piping.

Steam traps, including valving, are shown on the P&IDs and shall be detailed on the piping drawings.

9.4 Utility Station

Utility stations for utility steam, air and water and when required nitrogen shall be provided. Provide a hose rack at each station with 1 inch hose connections, the stations shall be located at grade or in structures platform with 15 m length of hose.

9.5 Vents and Drains

Unvalved Vent connections will be provided at high points in water and oil piping and high points in other piping, including steam lines, which will be hydrostatically tested. Unvalved Drain connections will be provided at non-drainable points in steam, water and oil piping and all other piping that will be hydrostatically tested.

Valve Drain connections will be provided for piping lines containing liquid at piping low points.

Only vent and drain connections that require frequent operation, or which may discharge significant quantities of fluid, will be piped to drain funnels. Vent or drain connections that normally require operation at a time when hot fluids are discharged will be piped to a safe termination point (drain funnel or floor area discharge).

The P&ID diagrams show the location of process vents, drains and bleeds used during plant operation. These connections are furnished with valve and shall be in accordance with relevant piping class unless otherwise indicated on the flow diagrams.

9.6 Pipe Branch Connections

Any line taken off from main line is considered a branch connection. Branch connections shall be made by the use of tees, reducing outlet tees, sockolets, sweepolets, reinforced stub-ins or non reinforced stub-ins.

Branch connections shall be made in accordance with the relevant tables shown in the piping material classes.

9.7 Piping class specifications

Code standard to be used for piping class will be: ASTM/ANSI B31.1.

9.8 Cathodic Protection

Where required, underground piping will be electrically isolated from the aboveground piping and from other steel components, to allow the underground piping to be cathodically protected. Isolation will be achieved by installation of isolation flanges with insulating gaskets, tubes and washers. Sacrificial anodes and/or impressed currents will be used for cathodic protection. The design of the system, materials, installation and testing will be in accordance with applicable standards.

9.9 Technical Consideration

Design, selection of construction materials, fabrication, testing and certification of all plant pipes, tubings, valves, fittings and pipe supports will conform to the following codes, as applicable, or approved equivalent:

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- ANSI/ASME B31.1 - Power Piping Code
- ASME Boiler and Pressure Vessel Code - Section I - Power Boilers
- ASME Boiler and Pressure Code - Section VIII, Division I - Pressure Vessels
- American Water Works Association (AWWA)

9.10 Other Facilities

The Turbine, Boiler, Stack are scope of vendor. Piping shall arrange for these systems in accordance with vendor packages tie-ins. Clear access road shall provide for trucks movement of ash handling system.